BEFORE: Pylint (2.41/10) and start of Bandit

radio_schedule.py | Lab 5: Static Code Analysis

```
radio_schedule.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 2.41/10

(venv) rachelmathew@Rachels-MacBook-Air L5_22_radio_schedule % clear
pylint radio_schedule.py

************* Module radio_schedule
radio_schedule.py:4:0: C0301: Line too long (118/100) (line-too-long)
radio_schedule.py:11:0: C0301: Line too long (135/100) (line-too-long)
radio_schedule.py:27:0: C0301: Line too long (110/100) (line-too-long)
radio_schedule.py:1:0: C0114: Missing module docstring (missing-module-docstring)
radio_schedule.py:6:0: C0103: Constant name "booked_minutes" doesn't conform to UPPER_CASE naming style (invalid-name)
radio_schedule.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
radio_schedule.py:9:0: C0103: Function name "addSlot" doesn't conform to snake_case naming style (invalid-name)
radio_schedule.py:10:4: W0603: Using the global statement (global-statement)
radio_schedule.py:11:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
radio_schedule.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
radio_schedule.py:15:0: C0103: Function name "slotLabel" doesn't conform to snake_case naming style (invalid-name)
radio_schedule.py:16:11: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
radio_schedule.py:19:0: C0116: Missing function or method docstring (missing-function-docstring)
radio_schedule.py:19:0: C0103: Function name "overlapFree" doesn't conform to snake_case naming style (invalid-name)
radio_schedule.py:22:11: W0718: Catching too general exception Exception (broad-exception-caught)
radio_schedule.py:21:15: W0123: Use of eval (eval-used)
radio_schedule.py:26:0: C0116: Missing function or method docstring (missing-function-docstring)
radio_schedule.py:26:0: C0103: Function name "morningShows" doesn't conform to snake_case naming style (invalid-name)
radio_schedule.py:27:27: C0321: More than one statement on a single line (multiple-statements)
radio_schedule.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
radio_schedule.py:31:0: C0103: Function name "removeSlot" doesn't conform to snake_case naming style (invalid-name)
radio_schedule.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)

------------------------------------------------------------------
Your code has been rated at 2.41/10 (previous run: 2.41/10, +0.00)

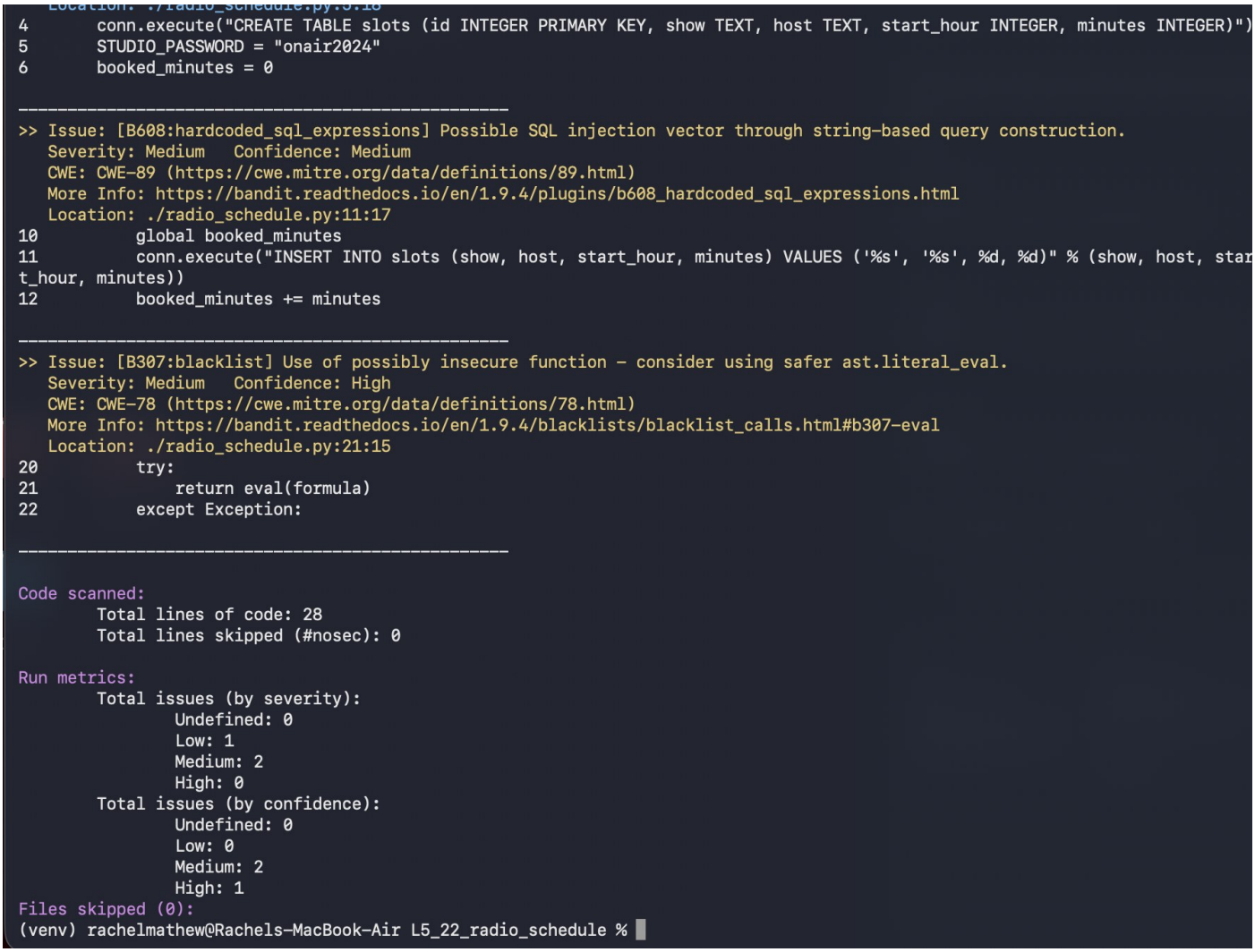
(venv) rachelmathew@Rachels-MacBook-Air L5_22_radio_schedule % clear
bandit -r radio_schedule.py

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.6
Run started:2026-10-18 06:21:33.751288+00:00

Test results:
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'onair2024'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: ./radio_schedule.py:5:18
```

BEFORE: Bandit (continued)

radio_schedule.py | Lab 5: Static Code Analysis



```
   Location: ./radio_schedule.py:3:18
4       conn.execute("CREATE TABLE slots (id INTEGER PRIMARY KEY, show TEXT, host TEXT, start_hour INTEGER, minutes INTEGER)")
5       STUDIO_PASSWORD = "onair2024"
6       booked_minutes = 0

--------------------------------------------------
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: ./radio_schedule.py:11:17
10          global booked_minutes
11          conn.execute("INSERT INTO slots (show, host, start_hour, minutes) VALUES ('%s', '%s', %d, %d)" % (show, host, star
t_hour, minutes))
12          booked_minutes += minutes

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: ./radio_schedule.py:21:15
20          try:
21              return eval(formula)
22          except Exception:

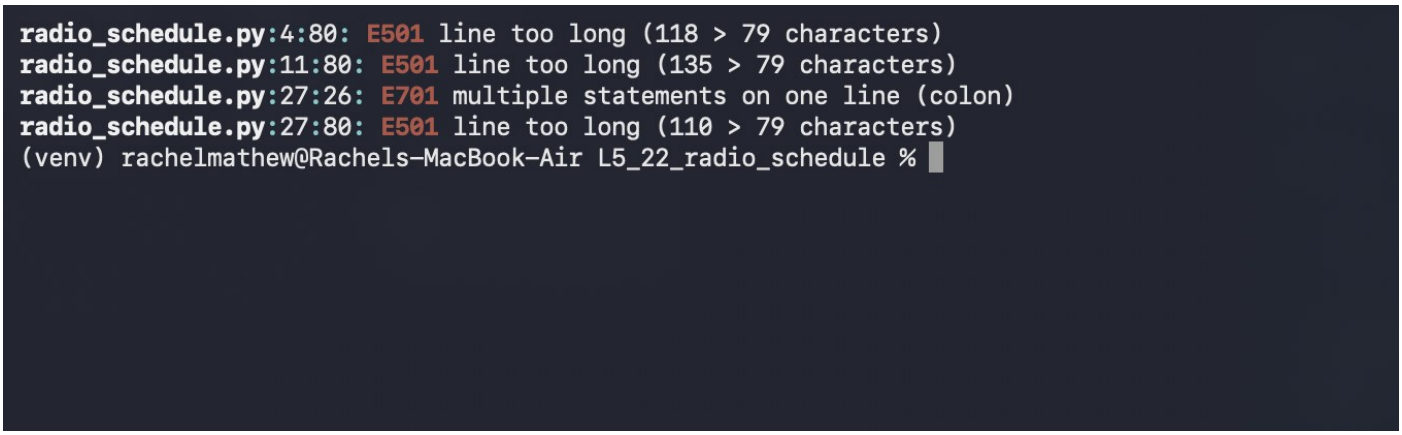
--------------------------------------------------

Code scanned:
        Total lines of code: 28
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 1
                Medium: 2
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 2
                High: 1
Files skipped (0):
(venv) rachelmathew@Rachels-MacBook-Air L5_22_radio_schedule %
```

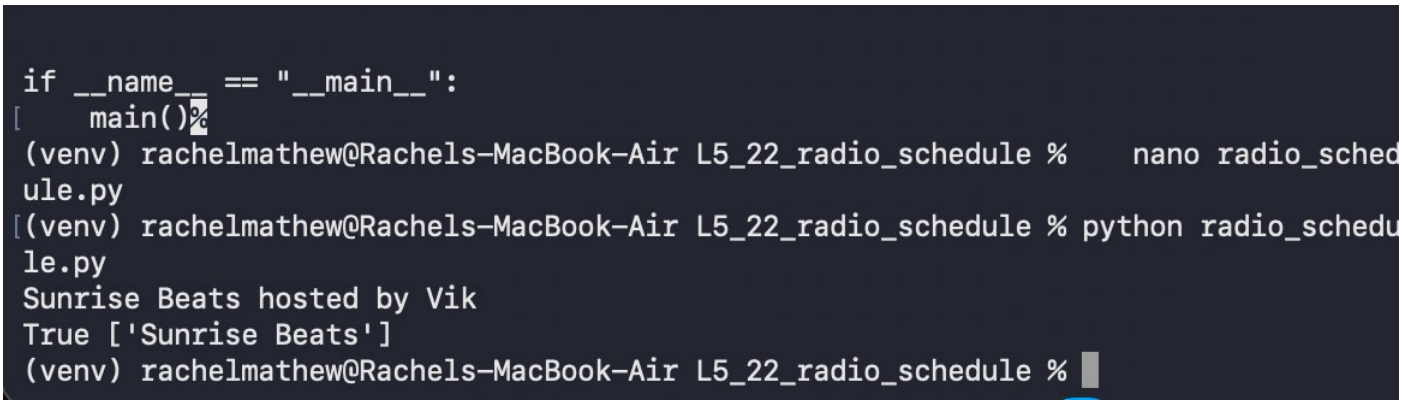
BEFORE: Flake8 (4 issues)

radio_schedule.py | Lab 5: Static Code Analysis



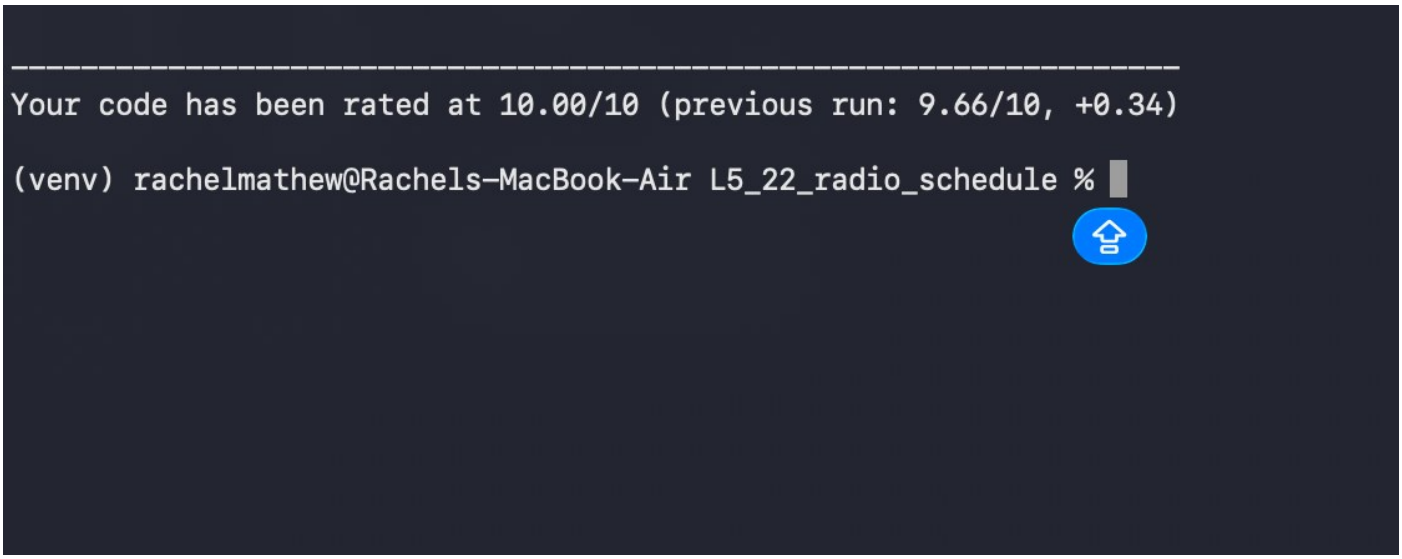
AFTER: Program still runs correctly

radio_schedule.py | Lab 5: Static Code Analysis



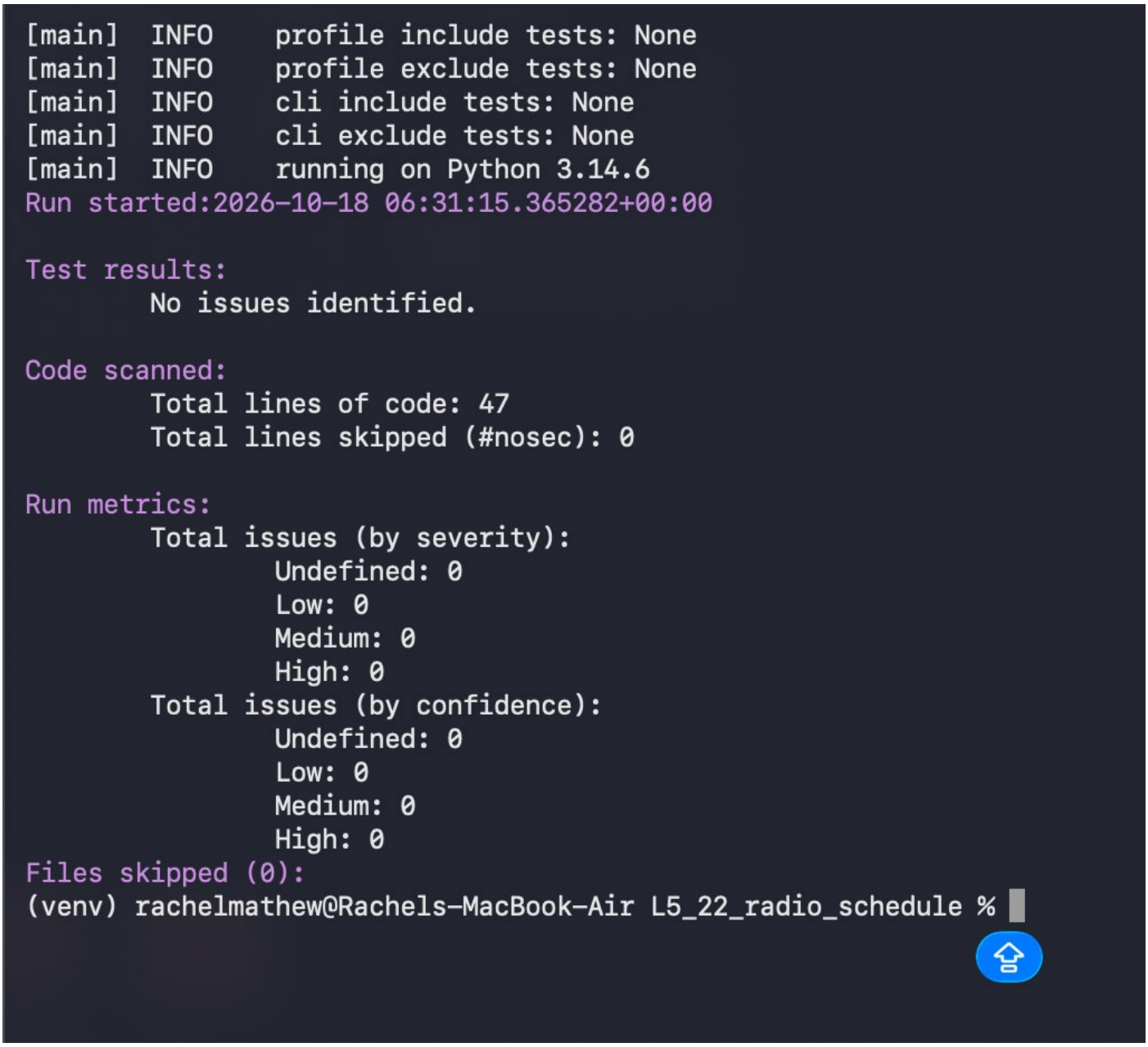
AFTER: Pylint (10.00/10)

radio_schedule.py | Lab 5: Static Code Analysis



AFTER: Bandit (no issues identified)

radio_schedule.py | Lab 5: Static Code Analysis



AFTER: Flake8 (exit code 0, no issues)

radio_schedule.py | Lab 5: Static Code Analysis

